

Experiment: 3

Experiment on Simulation Tool – Cisco Packet Tracer

Aim

To create a simple computer network using Cisco Packet Tracer, configure IP addresses for the devices, test connectivity using the ping command, and observe packet transmission in simulation mode.

Software Required

- Cisco Packet Tracer
- Windows operating system
- Computer or Laptop

Procedure

1. Open Cisco Packet Tracer.
2. Create a new workspace.
3. Place two PCs and one switch in the workspace.
4. Connect the PCs to the switch using appropriate cables.
5. Configure IP addresses for both PCs.
6. Test the connection using the ping command.
7. Switch to Simulation Mode to observe packet transmission.

Topology Setup

The network topology consists of two computers connected through a switch.

Topology: PC0 → Switch → PC1

Devices used: 2 PCs (PC0 and PC1) and 1 Switch (2960).

Cable Connection

Device 1	Port	Device 2	Port
PC0	FastEthernet0	Switch	FastEthernet0/1
PC1	FastEthernet0	Switch	FastEthernet0/2

Use Copper Straight-Through Cable for the connections.

IP Configuration

Device	IP Address	Subnet Mask
PC0	192.168.1.1	255.255.255.0
PC1	192.168.1.2	255.255.255.0

For each PC: Click the PC → Desktop → IP Configuration → Enter the IP address and subnet mask.

Testing

1. Click PC0.
2. Open Desktop → Command Prompt.
3. Type:

```
ping 192.168.1.2
```

Expected result: Reply from 192.168.1.2. Successful replies indicate that the network is properly configured.

Simulation

1. Click Simulation Mode at the bottom-right corner of Cisco Packet Tracer.
2. Select Add Simple PDU (envelope icon).
3. Click PC0 as the source.
4. Click PC1 as the destination.
5. Click Auto Capture/Play.
6. Observe the packet moving from PC0 → Switch → PC1.

The simulation shows packet transmission between the devices and confirms successful communication.

Observation

The two PCs were successfully connected through a switch using copper straight-through cables. IP addresses were assigned within the same subnet. The ping command produced successful replies, and Simulation Mode showed the packet traveling from PC0 through the switch to PC1.

Result

Thus, a basic computer network was successfully created using Cisco Packet Tracer. The devices were configured with valid IP addresses, connectivity was verified using the ping command, and packet transmission was observed successfully in Simulation Mode.